Machine Learning

A Beginner’s Guide for a PDF Chatbot (RAG) Project

A compact learning document covering core machine-learning ideas, common algorithms, model evaluation, and practical terminology. Use it as the knowledge document for your PDF question-answering chatbot.

Learning level: Beginner
Suggested use: Upload this PDF into your RAG application, then ask questions about its sections.

Prepared as an original study resource. It is not a published textbook.

1. What is Machine Learning?

Machine Learning (ML) is a branch of artificial intelligence in which computers learn patterns from data and use those patterns to make predictions or decisions. Instead of writing every rule manually, a developer provides examples and a learning algorithm finds useful relationships.

A machine-learning system commonly includes data, features, a learning algorithm, a trained model, and predictions. For example, an email spam classifier learns from messages labelled “spam” and “not spam.” After training, it predicts the class of a new email.

Key points

- Artificial Intelligence (AI) is the broader field of building systems that perform tasks associated with intelligence.
- Machine Learning is one approach within AI that learns from data.
- Deep Learning is a subset of ML that uses multi-layer neural networks.

2. Main Types of Machine Learning

Supervised learning uses labelled examples. Each training example contains input features and a known target. The model learns to predict the target for new inputs.

Unsupervised learning uses data without target labels. The algorithm searches for structure, such as groups of similar customers.

Reinforcement learning trains an agent to choose actions by interacting with an environment. The agent receives rewards or penalties and learns a strategy that aims to maximize long-term reward.

Key points

- Supervised example: predict whether a transaction is fraudulent using labelled historical transactions.
- Unsupervised example: group customers by purchasing behaviour using clustering.
- Reinforcement-learning example: train a game-playing agent through rewards for successful moves.

3. Features, Labels, Training Data and Testing Data

A feature is an input variable used by a model. In a house-price model, examples include area, number of bedrooms, and location. A label (or target) is the value the model is expected to predict, such as the house price.

The training set is used to fit the model. The test set is held back until evaluation so that performance is measured on examples the model did not train on. A validation set can be used to compare model settings during development.

Data leakage occurs when information unavailable at prediction time enters training. Leakage can make evaluation scores look unrealistically high.

Key points

- For churn prediction, features might include purchase frequency and days since last purchase.
- The label might be whether the customer churned.
- Split data before fitting preprocessing steps to avoid information leaking from the test set.

4. Common Supervised Learning Tasks

Classification predicts a category. Examples include spam versus not spam, fake versus factual news, and positive versus negative sentiment.

Regression predicts a numeric value. Examples include house prices, delivery time, and monthly sales.

A classification model may return class probabilities as well as a predicted class. A probability is a model estimate, not a guarantee.

Key points

- Binary classification has two classes.
- Multiclass classification has more than two possible classes.
- Regression outputs a continuous numeric prediction.

5. Common Machine Learning Algorithms

Linear Regression models a numeric target as a weighted combination of input features. It is often used as a simple regression baseline.

Logistic Regression is commonly used for classification. Despite its name, it estimates class probabilities using a logistic function.

Decision Trees make predictions through a sequence of feature-based questions. They are relatively easy to interpret but can overfit if allowed to grow too deep.

Random Forest combines predictions from many decision trees trained with randomness. It often improves stability and predictive performance compared with a single tree.

K-Nearest Neighbours (KNN) predicts using the labels or values of nearby training examples. The definition of “nearby” depends on a distance measure and feature scaling.

Support Vector Machines (SVMs) find a decision boundary that separates classes, with the option of using kernels for nonlinear boundaries.

K-Means is an unsupervised clustering algorithm that assigns examples to a chosen number of clusters.

Key points

- Choose algorithms based on the problem, dataset size, interpretability needs, and evaluation results.
- Always compare a more complex model with a simple baseline.

6. Data Preprocessing

Real-world data often needs cleaning before it can be used. Common steps include handling missing values, removing or investigating duplicate records, correcting inconsistent formats, and encoding categorical variables.

Numerical features may need scaling. Standardization transforms a feature so it has approximately zero mean and unit variance. Scaling is particularly important for distance-based methods such as KNN and for many optimization-based models.

Text preprocessing may include normalization, tokenization, stop-word handling, and lemmatization. The appropriate steps depend on the task and model; modern transformer models often use their own tokenizer and generally should not receive aggressive manual preprocessing.

Key points

- Fit scalers and encoders on training data only, then apply the fitted transformations to validation and test data.
- Do not delete duplicate rows automatically without checking whether they are genuinely redundant.

7. Overfitting and Underfitting

Overfitting happens when a model learns details and noise specific to the training data. It performs very well on training examples but poorly on unseen data.

Underfitting happens when a model is too simple or insufficiently trained to capture the underlying pattern. It performs poorly on both training and test data.

Ways to reduce overfitting include collecting more representative data, using cross-validation, simplifying the model, applying regularization, pruning trees, and using early stopping where appropriate.

Key points

- A large gap between training and test performance can be a warning sign of overfitting.
- Poor performance on both sets can indicate underfitting, weak features, or data-quality problems.

8. Model Evaluation Metrics

Accuracy is the fraction of all predictions that are correct. It can be misleading when classes are imbalanced.

Precision answers: of the examples predicted positive, how many were actually positive?

Recall answers: of all actual positive examples, how many did the model find?

The F1-score is the harmonic mean of precision and recall. It is useful when both false positives and false negatives matter.

A confusion matrix counts true positives, true negatives, false positives, and false negatives. For regression, common metrics include mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE).

Key points

- Select metrics based on the cost of different errors.
- For rare fraud cases, recall and precision may be more informative than accuracy alone.

9. The Typical Machine Learning Workflow

A practical ML workflow starts by defining the problem and the target. Next, collect and inspect data, clean it, explore patterns, and split it into training and test sets.

Build a simple baseline, train candidate models, tune hyperparameters using training data and cross-validation, and evaluate the selected approach on held-out test data. Finally, save the preprocessing pipeline and model, deploy them, monitor their performance, and retrain when data or behaviour changes.

A model is not finished when training ends. Production systems need monitoring for data drift, performance degradation, latency, and unexpected failures.

Key points

- Problem definition → data collection → preprocessing → train/test split → training → validation/tuning → testing → deployment → monitoring.

10. Machine Learning and a RAG PDF Chatbot

A Retrieval-Augmented Generation (RAG) chatbot answers questions using information retrieved from a knowledge source. In a PDF chatbot, the PDF is loaded, its text is split into chunks, and each chunk is converted into an embedding (a vector representation of meaning).

The vectors and their associated text are stored in a vector database such as ChromaDB or FAISS. When a user asks a question, the system embeds the question, searches for similar chunks, and sends the retrieved context to a language model to generate an answer.

RAG is not the same as training a machine-learning model on the PDF. The document is indexed for retrieval; the language model uses the retrieved passages as context when answering.

Key points

- Document loading: extract text from PDF pages.
- Chunking: divide long text into smaller overlapping sections.
- Embedding: map text into numerical vectors.
- Vector search: retrieve chunks semantically similar to the question.
- Generation: produce an answer grounded in retrieved context.

Glossary

Algorithm: A procedure or set of rules used to learn patterns or make predictions.

Model: The learned function or system produced after training.

Embedding: A numerical vector representation of text or another data item.

Hyperparameter: A setting chosen before or during model training, such as tree depth or learning rate.

Cross-validation: A method that evaluates a model across multiple train/validation splits.

Data drift: A change over time in the distribution of model input data.

Vector database: A system designed to store and search vector representations efficiently.

Similarity search: Retrieval of items whose vector representations are close to a query vector.

Practice questions for your chatbot

1. What is the difference between supervised and unsupervised learning?
2. How are classification and regression different?
3. What is the purpose of a training set and a test set?
4. Explain overfitting in simple terms.
5. What is the difference between precision and recall?
6. Why should a scaler be fitted only on training data?
7. How does a Random Forest make predictions?
8. What is an embedding in a RAG application?
9. How does a vector database help a PDF chatbot?
10. Does RAG train the language model on the uploaded PDF?